

Nautatava Navlakha

Indian Institute of Technology Bombay, 2020

B.Tech, Civil Engineering

Minor, Computer Science

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Extremely savvy in building automation tools, writing high-quality code and mentoring people

- Managed 5+ clients to ensure timely delivery, often catering to more than 2 clients at a time
- Built **10+ automation tools** and plugins, both proprietary and open-source
- Involved in **15+ teams on 30+ projects** covering on-ground as well as management positions
- **Mentored 50+ people** and delivered seminars for educating colleagues for new developments

Work Experience

Consultant | FinMechanics India Pvt. Ltd.

[Jul'20 – Present]

Automating cross-asset treasury platform for trade pricing, risk, compliance and processing

• Management

- Conceptualized and led a team of 4 implementation engineers to develop **automation pipelines** impacting **50+ clients** and **10000+ people**, introducing the concept of **dev-sec-ops** in project delivery workflow
- Person in charge of implementation and go-live of fixed-income back-office modules on *fmconverge* to handle and support portfolio value of **INR 75,000+ Crore** for one of the largest bank FI debt markets in India
- Demonstrated and initiated discussions over tech stack used and infrastructure needed to handle client load

• Technical

- Optimized the core operations of *fmconverge* by over 1000% allowing deals to be processed in less than a second against the previous 5 seconds by **vectorized pricing** and **Asynchronous non-blocking programming**
- Added support for multiple authentication protocols implementing LDAP non-anonymous bind, SSO and 2FA

• Algorithm Development

- Designed fuzzy logic algorithm to pin down most appropriate payment method from user-defined choices
- Architected a **new sorting engine** to efficiently sort end reports for users based on given set of features
- Developed regex/JS based number-range inputs for faster navigation from 3.5M+ deals booked on system

Co-Founder and Chief Technical Officer | Kwipic.in

[Mar'20 – Oct'21]

Building low cost, no-bias accurate CV models

• Superintendence

- Oversaw development of **no-bias single-shot face recognition algorithm** learned over 2 million+ faces
- Spearheaded the automation of recognition engine, saving **600 employee hours** and **₹3L per month**
- Coordinated the adoption of **agile methodology** - often going from idea to deploying the same day!

• Implementation

- Automated complete application workflow using **event-driven architecture** and improved performance by over 10 times by implementing **parallel computing**, **selective caching** and **multi-threading**
- Delivered 75k+ photos to 5k+ users using face recognition and expanded the overall client support by integrating Google Drive, DropBox and OneDrive APIs into content delivery workflow
- Native Support for **authorizing APIs at origin** instead of filtering the requests in logic layer to remove human mistakes in development and **improve app-security**

• Achievements and milestones

- Built the complete platform from PoC to scalable production-grade system to a final commercial release
- Serviced 100+ events and partnered with 30+ photographers to expand reach and create market space

Internships

Summer Associate | FinMechanics India Pvt. Ltd.

[May'19 – Jul'19]

Automating cross-asset treasury platform for trade pricing, risk, compliance and processing

- Received **Pre-Placement Offer** for exceptional performance over the course of internship as summer associate
- Created **self-updating views** for real-time data visualization and better UCM integration on production
- Completely revamped the core part of product - “bond master engine” to better abstract 105 feature groups
- Developed APIs in **Java Spring**, DB abstracted by **Hibernate**, **Hazelcast** for in-memory caching, **ExtJS** front-end

MERN Stack Developer | Skillate Industries Pvt. Ltd.

[May'18 – Jul'18]

A job-matching platform for enterprises to make the recruitment process hassle-free using artificial intelligence

- Implemented **single page progressive** web application using React JS which fetches data from API dynamically
- Integrated Google Analytics, Google Jobs and SEO into the web-app to increase the web-reach of application
- Used **Server-Side Rendering** for faster loading times, **service-worker** for offline functioning of job applications
- Built Landing page from scratch using ExpressJS, Jade Templating Engine with **sprites** for better performance

SEO analyst | Toppr Learning

[Dec'17 – Jan'18]

An online practice platform helping students prepare for IIT JEE, PMT, school boards and other competitive exams

- Studied keywords, competitors, backlinks, content using tools like **Google Webmasters**, **Ahrefs** and **SEOquake** to implement effective SEO solutions from scratch; also ensured, changes are worked upon by the content team
- Measures taken on analysis led to an increase in as much as **50% organic traffic** on webpages in 20 days

Technical Skills

- **Programming Languages** Java (OOP, Android Studio), C/C++, SQL, Python, Bash
- **Web Development** HTML, CSS, JavaScript (Vanilla, Angular, React, Express, Ext), Pug
- **Frameworks** Spring, Hibernate, Django, SQL Alchemy, Flask, Elastic Search
- **Databases** SQL server, MySQL, PostgreSQL, Redis, MongoDB
- **Tools** Git, Docker, JIRA, $\text{\LaTeX}$

Technical Achievements

Open-source contributions

- Recipient of **Arctic Code Vault Contributor** for special developments in [Caprine](#), [Django-robots](#) and [webgems](#)
- Awarded on issuehunt for long awaited **spell-check implementation** for applications based on [electron-context-menu](#) (**4000+ dependent repositories** and **75+ packages**)

Institute System Administrator (ISA)

Top-ranked technical role at IIT Bombay

- Awarded with **Institute Organizational Commendation** for exceptional work as ISA
- Developed **Single-Sign-On architecture** following **OIDC/SAML** standards for common login across all the services
- Built [iOS client](#) and [dev-friendly Django plugin](#) for secure **user-initiated password-less LDAP authentication**
- Administered the [student datacenter](#) hardware infrastructure and migrated from virtualization to LXC
- Made **AMS** - a **cryptographically secure solution** to digitize the decade-old paper-based workflows as an emergency response to COVID-19

Research

- Successfully defended research on **impact of super-resolution** on high-quality **road detection** by mid-resolution satellite images. Results show correct road detection of 90% on 2-lane roads compared with state-of-the-art 73%
- Worked on data processing and neural net architecture to use SR on spatial road detection with **deep learning**

Extra-curriculars

Mentorship	• Responsible for knowledge transfer and handover for 2 interns coming as summer associate
	• Received with highest praise, Institute mentorship Special mention 2020
	• Led the way for 12 Juniors and 3 Academically underperforming students
	• Guided 25 freshmen and maintained reports about complete work done as part of NSS IIT Bombay
Seminars	• <i>Best practices when coding and it's benefits</i> - FinMechanics
	• <i>Knowledge transfer about development of new modules</i> - FinMechanics
	• <i>Getting started on technical stuff</i> - 1000+ students, freshmen induction program - IIT Bombay
	• <i>Teaching the under-privileged</i> - 5 NGOs and 200+ students - National Service Scheme
Social work	• Co-organised "CURED"- PAN India Social Initiative spreading diabetes awareness in 50+ cities
	• Completed 100+ hours of community service in NGOs as an NSS volunteer
	• Built a just-in-time inventory system to improve logistics and reduce overall costs by 50%. Led a team of 25 field officers, efficiently managing on-ground activities for driving social initiatives